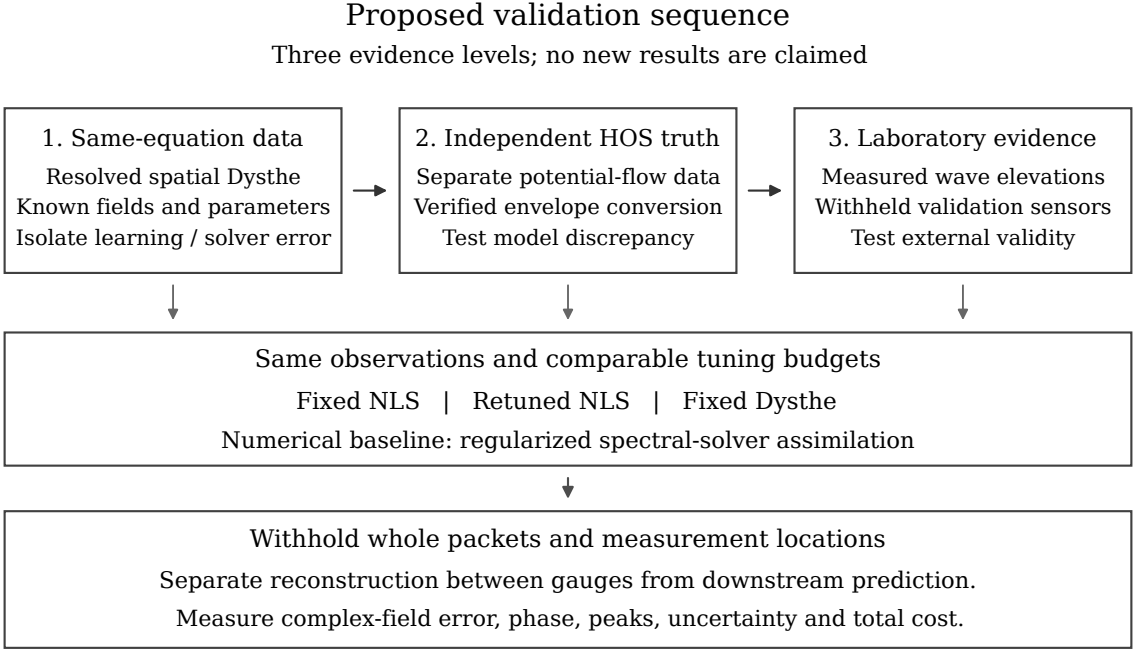
Proposed validation sequence
Three evidence levels; no new results are claimed
1. Same-equation data
Resolved spatial Dysthe
Known fields and parameters
Isolate learning / solver error
2. Independent HOS truth
Separate potential-flow data
Verified envelope conversion
Test model discrepancy
3. Laboratory evidence
Measured wave elevations
Withheld validation sensors
Test external validity
Same observations and comparable tuning budgets
Fixed NLS | Retuned NLS | Fixed Dysthe
Numerical baseline: regularized spectral-solver assimilation
Withhold whole packets and measurement locations
Separate reconstruction between gauges from downstream prediction.
Measure complex-field error, phase, peaks, uncertainty and total cost.